



SENTINEL

1.0x

COMPOUND THROUGH THE NOISE.

A deterministic Wealth Core risk architecture

Engineering specification • research prospectus • validation dossier

22.12%

20Y CAGR

-23.93%

MAX DD

54.42x

\$1 ENDING MULTIPLE

Exact next-open Sharadar replay • July 31, 2006 - July 31, 2026

01 • THE PROPOSITION



Executive overview

Sentinel 1.0x is the current research champion built around certified Wealth Core. Its central rule is deliberately simple: ordinary stress is a sensor, not an actuator. The portfolio remains fully exposed through ordinary corrections and moves defensive only when fast systemic or slow persistent evidence establishes severe stress.

This document is intended to be detailed enough to implement, replay, audit and certify the complete system. It separates the Wealth Core alpha engine, the immutable shadow control plane, the Sentinel risk overlay and the exact next-open execution layer.

Measure	Sentinel 1.0x	Strategy 6	SPY
20Y CAGR	22.12%	20.42%	11.26%
Max drawdown	-23.93%	-28.01%	-55.20%
Ending multiple	54.42x	41.08x	8.44x
Overlay transitions	14	43	n/a

Design principle: own durable winners, tolerate normal corrections, and reserve portfolio-level defense for evidence of genuine severe stress.

02 • ONE ALPHA ENGINE, ONE SHADOW, ONE SEVERE-STATE ACTUATOR



System architecture

Layer 1 - Wealth Core: a stateful 25-slot durable-winner engine. Layer 2 - Immutable shadow: a full-exposure counterfactual Wealth Core book. Layer 3 - Sentinel: a deterministic risk controller. Layer 4 - Execution: decisions after close t, fills at the next session open.

State / condition	Live Core	Defensive	Purpose
NORMAL	100%	0%	Full participation
ORDINARY_STRESS	100%	0%	Warning + persistence clock
SEVERE_STRESS / fast	0%	100%	Rapid systemic shock
SEVERE_STRESS / slow	0%	100%	Sustained negative expectancy

Canonical simplification: one severe-stress actuator with two independent evidence paths. The cause remains recorded because recovery semantics differ. This representation was proven allocation- and NAV-equivalent to the prior naming.

03 • FROZEN STOCK-SELECTION AND OWNERSHIP BEHAVIOR



Wealth Core alpha engine

Signal mathematics consume split-adjusted, dividend-unadjusted closes. Raw/as-traded prices are used for execution accounting. A candidate requires 127 closes spanning t-126 through t.

Component	Canonical rule
Medium-term momentum	$P[t-21] / P[t-126] - 1$
Leadership population	$\max(25, \text{ceil}(\text{eligible} \times 10\%))$
Freshness	$P[t] / P[t-21] - 1 \geq 0$
Formation volatility	$\text{sample stdev}(\log \text{ returns}) \times \sqrt{252}$
Durable score	$\ln(1 + \text{momentum}) / \text{volatility}$

Leadership membership is decided by raw medium-term momentum. Durable score is the final admission ordering within the qualified set. The certified volatility profile uses one-session log returns, not simple returns.

Portfolio construction: 25 slots, 4% nominal entry weight, whole shares, no leverage, 10 bps one-way transaction cost. Initial construction may fill all available slots; after initialization, at most one new admission is scheduled per session.

Admission exclusions: no duplicate held/pending permanent security or issuer; respect 21-session slot and security cooldowns; block new admissions if current equity cannot be resolved; include transaction cost in affordability. Aggregate guardrails are 25% per security and 30% per issuer.

04 • WINNERS ARE ALLOWED TO REMAIN WINNERS



Position lifecycle

Trailing stop: each holding maintains an episode peak from split-adjusted owned closes. Exit is triggered inclusively when current close $\leq 70\%$ of that peak, i.e. a 30% trailing stop. The peak ratchets upward only and is not seeded from the entry open.

One-time review: after 119 completed market sessions, exit only when current close is strictly below the split-adjusted entry price AND the security no longer passes full entry qualification. Otherwise mark the review permanently complete. Missing close means defer the review.

Event	Rule	Action
Entry	Leadership + freshness + durable rank	Buy next open
Normal holding	No rebalance / no rank churn	Hold
Trailing stop	Close $\leq 0.70 \times$ peak	Sell next open
119-session review	Underwater AND no longer qualifies	Sell next open
Review passes	Not both conditions	Never review again
Exit	Slot + security cooldown	21 sessions

05 • ORDINARY STRESS WATCHES; SEVERE STRESS ACTS



Sentinel 1.0x controller

Ordinary stress activates when immutable-shadow drawdown reaches -15.5% or worse. Sentinel does not reduce allocation. It records a stress anchor and starts the persistence clock.

```
severe_stress = fast_severe_trigger OR slow_severe_trigger  
target_core_allocation = 0.0 if severe_stress else 1.0  
# ordinary_stress changes memory, not allocation
```

Slow persistent entry	Rule
Ordinary stress	active
Stress age	≥ 30 sessions
Return since stress anchor	$\leq -2\%$
Shadow 40-session return	$\leq -3\%$
Damaged breadth	$\geq 75\%$
Green breadth	$\leq 25\%$

Slow recovery: remain severe for at least 20 sessions, then require five consecutive healthy confirmations: shadow r20 > 0, damaged breadth $\leq 60\%$, green breadth $\geq 20\%$.

06 • CONFIRMED SHOCK PATH



Fast systemic detector

The research-harness fast detector is conjunctive: drawdown alone is insufficient. It requires a broad, accelerating shock with short-horizon loss, breadth damage, volatility acceleration and market confirmation.

Feature	Research-harness condition
Shadow drawdown	$\leq -10\%$
Damaged breadth	$\geq 85\%$
Green breadth	$\leq 20\%$
Short loss	$r5 \leq -5\%$ OR $r10 \leq -8\%$
Damage acceleration	5-session damaged increase ≥ 40 pp
Volatility acceleration	$vol5 / vol20 - 1 \geq 4\%$
Confirmation	SPY $r20 \leq -1\%$ OR shadow $r10 \leq -10\%$

Fast recovery in the research harness: minimum 10 severe-state sessions plus three consecutive healthy confirmations. Re-arm after shadow drawdown improves above -6% and the shock signal is absent.

Implementation gate: import the frozen certified systemic configuration when Sentinel is coded into Stocker. If the certified source differs from this research-harness definition, the source configuration wins and the strategy version changes.

07 • THE STRATEGY'S COUNTERFACTUAL TRUTH



Immutable shadow control plane

The shadow is not a benchmark. It is a complete counterfactual Wealth Core book that remains fully exposed regardless of live defense. Every event that would affect full-exposure Wealth Core must continue in the shadow.

Required state	Examples
Portfolio	cash, positions, shares, marks, NAV, high-water
Holding episode	permanent ID, entry, peak, review, age
Control	slot/security cooldowns, reservations, pending orders
Corporate actions	receivables, conversions, terminal/settlement state
Derived risk	DD, r5/r10/r20/r40, damaged/green, vol acceleration
Audit	state version, lineage hash, decision hash, timestamps

Prohibited shortcuts: do not rebuild the shadow from current rankings after defense; do not reset entry dates or peaks; do not use live NAV for shadow drawdown; do not stop applying dividends, splits, mergers, delistings, exits or admissions while live is defensive.

08 • CLOSE DECIDES; NEXT OPEN EXECUTES



Daily operating cycle

TRADING SESSION t CLOSE

- 1 Finalize point-in-time market / corporate-action inputs
- 2 Advance live Wealth Core and immutable shadow
- 3 Mark positions; apply dividends and corporate actions
- 4 Evaluate position exits and rank admissions
- 5 Compute shadow risk features
- 6 Update ordinary-stress memory
- 7 Evaluate fast severe trigger
- 8 Evaluate slow severe trigger / recovery
- 9 Resolve target Core allocation: 100% or 0%
- 10 Persist shadow + controller + decision atomically
- 11 Enqueue next-open actions

TRADING SESSION $t+1$ OPEN

- 12 Mark old book at actual open
- 13 Execute pending Wealth Core exits / entries
- 14 Execute changed Core / defensive notional
- 15 Apply costs and persist fills atomically

No-look-ahead invariant: session t 's decision may use only information finalized at or before t close. The $t+1$ open is execution data. Deployment uses a 252-session warm-up, comfortably covering the 126-session signal requirement and controller features.

09 • EXACT NEXT-OPEN SEMANTICS



Execution and accounting

For a transition decided after close t and executed at open $t+1$, the old allocation owns the overnight gap and the new allocation owns the open-to-close move. Existing Wealth Core positions are priced at the transition open before changing exposure.

```
overnight = old_alloc * WC(close_t -> open_t+1)
           + (1-old_alloc) * TBILL(close_t -> open_t+1)

intraday  = new_alloc * WC(open_t+1 -> close_t+1)
           + (1-new_alloc) * TBILL(open_t+1 -> close_t+1)

cost      = one_way_cost * abs(new_alloc-old_alloc)
return    = (1+overnight)*(1+intraday)*(1-cost)-1
```

Research overlay transition cost is 10 bps one-way on changed notional. The defensive sleeve is a short-duration Treasury-bill total-return proxy; BIL was used in the validated overlay. Production mapping to BIL, SGOV or direct T-bills must explicitly define distributions, settlement, liquidity and taxes.

10 • RESTART MUST NOT CHANGE THE ANSWER



Persistence and data contracts

Field	Requirement
controller_state_version	immutable version
as_of_session	trading-session date
ordinary_stress + anchor	boolean + date/NAV
severe_stress	boolean
fast / slow cause	separate booleans
state ages / healthy streaks	trading-session counters
last / pending allocation	0.0 or 1.0
feature snapshot	all decision inputs
reason codes	ordered immutable list
input/config/code hashes	replay identity

Atomicity: persist shadow state, controller state and decision before submitting orders. At the next open, persist fills and resulting live state atomically. Missing critical prices follow an explicit fill policy and are logged. Unresolved equity blocks admissions but never risk-reducing exits.

11 • CODIFICATION BLUEPRINT



Reference pseudocode

```
def on_close(session):
    inputs = point_in_time_inputs(session)
    shadow = wealth_core_step(shadow, inputs)
    live = wealth_core_step_live(live, inputs)
    f = compute_shadow_features(shadow)

    ordinary = update_ordinary_memory(f.dd, threshold=-0.155)
    fast = update_fast_severe(controller.fast, f)
    slow = update_slow_severe(controller.slow, ordinary, f)

    severe = fast.active or slow.active
    target = 0.0 if severe else 1.0

    decision = Decision(session, target, ordinary, fast, slow, f, hashes())
    atomic_persist(shadow, live, controller, decision)
    enqueue_next_open_actions(decision)

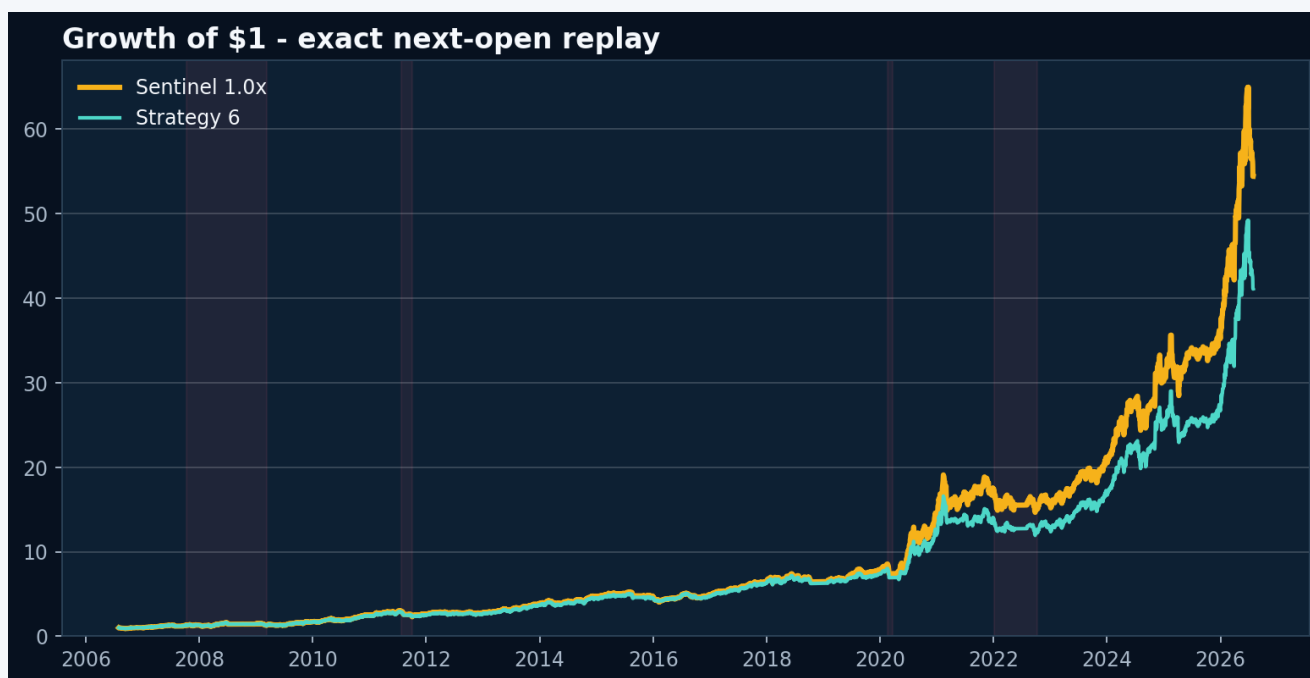
def on_next_open(session):
    decision = require_prior_close_decision(session)
    mark_old_book_at_actual_open()
    execute_pending_wealth_core_orders()
    execute_changed_core_notional(decision.target)
    execute_defensive_sleeve_complement()
    charge_costs()
    atomic_persist_fills_and_live_state()
```

Determinism requirements: ages count eligible trading sessions; threshold inclusivity is frozen; holdings are sorted by permanent security ID before breadth aggregation; configuration and ordering profiles are hashed; no random tie-breaking.

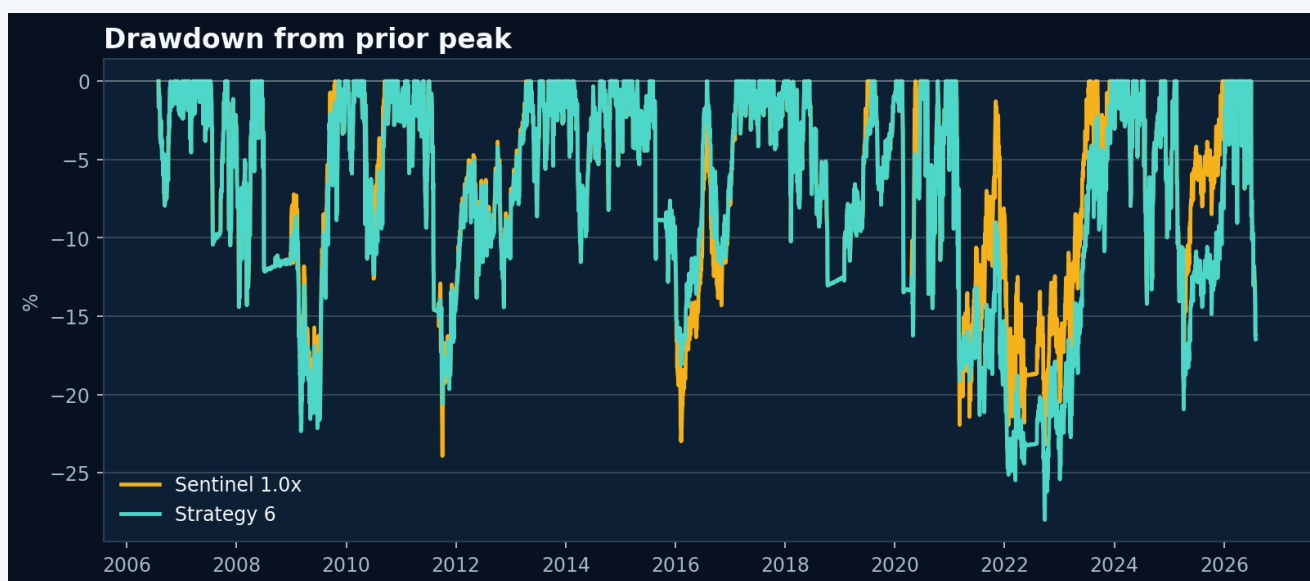
12 • EXACT NEXT-OPEN REPLAY



Historical performance



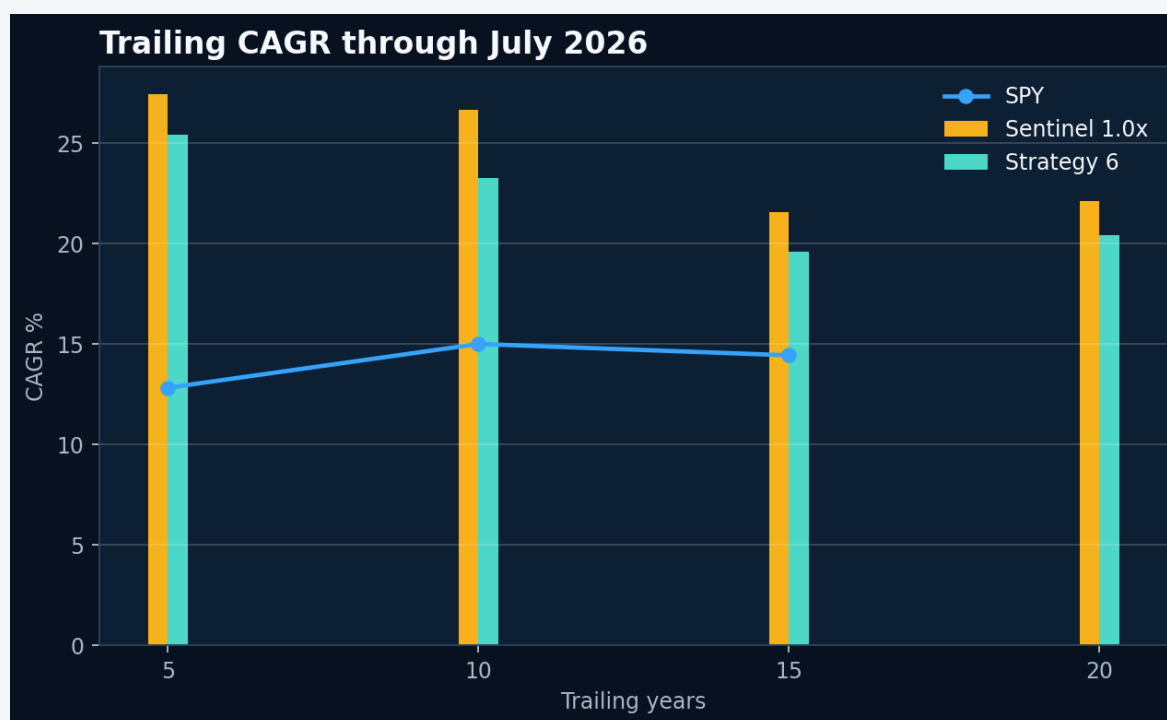
Sentinel recorded only 14 overlay transitions versus 43 for Strategy 6. The historical edge is therefore associated with less tactical portfolio-level trading, not more.



13 • THE EDGE IS NOT ONE ENDPOINT



Trailing and crisis behavior



Period	Sentinel CAGR	Sentinel DD	S6 CAGR	S6 DD
5Y	27.47%	-22.23%	25.41%	-20.95%
10Y	26.68%	-23.23%	23.25%	-28.01%
15Y	21.59%	-23.23%	19.62%	-28.01%
20Y	22.12%	-23.93%	20.42%	-28.01%

Crisis	Sentinel ret	Sentinel DD	S6 ret	S6 DD
2011	-15.2%	-23.9%	-15.5%	-20.6%
2022 Bear	-9.6%	-15.6%	-8.0%	-14.2%
COVID	23.5%	-13.8%	14.4%	-16.3%
GFC	10.9%	-21.2%	9.3%	-22.3%
Q4 2018	-4.5%	-8.3%	-4.5%	-8.3%

14 • DEPTH AND TIME UNDERWATER



Drawdown quality

Strategy	Worst DD	Days <= -10%	Days <= -20%	Days <= -25%	1% tail
Strategy 6	-28.01%	1,635	301	15	-23.88%
Sentinel	-23.93%	1,387	69	0	-20.38%

Historically, Sentinel spent 69 sessions at or below -20% drawdown versus 301 for Strategy 6, and no measured sessions below -25%. This is a historical property, not a guarantee: hostile slow-bear worlds show Sentinel can accept deeper drawdowns.

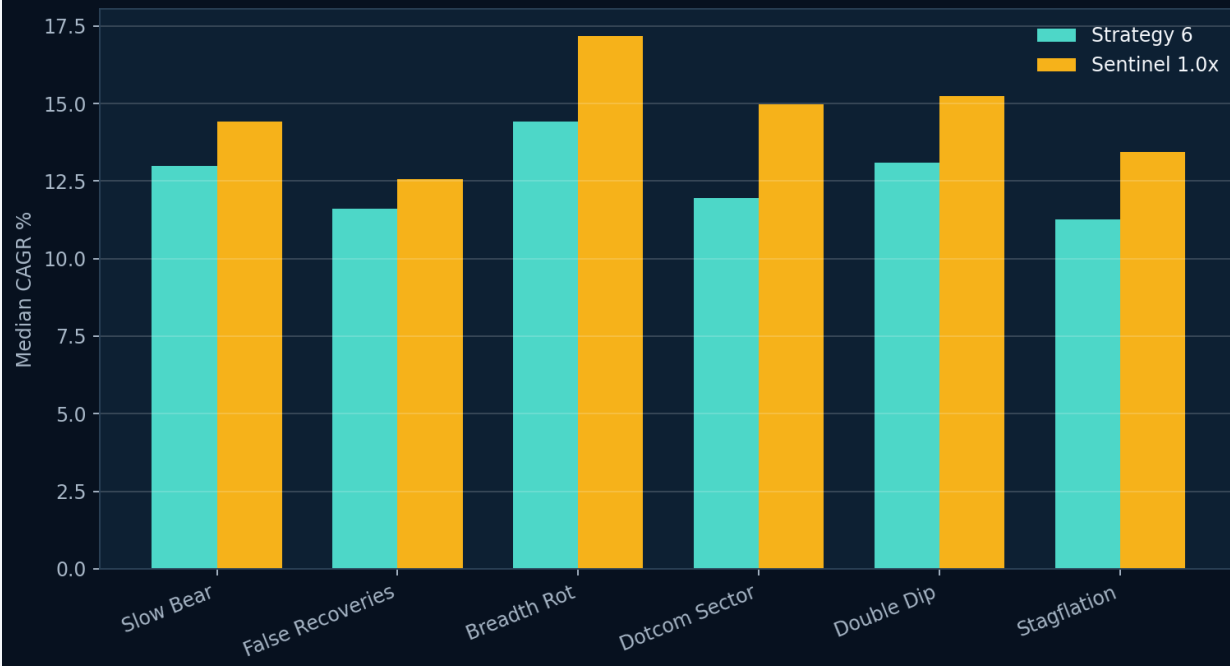
Risk philosophy: temporary volatility is tolerated when the strategy's long-run expectancy remains intact. Severe defense is reserved for evidence that the strategy itself is broadly failing.

15 • DELIBERATELY HOSTILE MARKET STRUCTURES



Adversarial testing

Adversarial worlds - return trade-off



Adversarial worlds - the price of less insurance



16 • SENTINEL PAYS LESS FOR INSURANCE



What the adversaries say

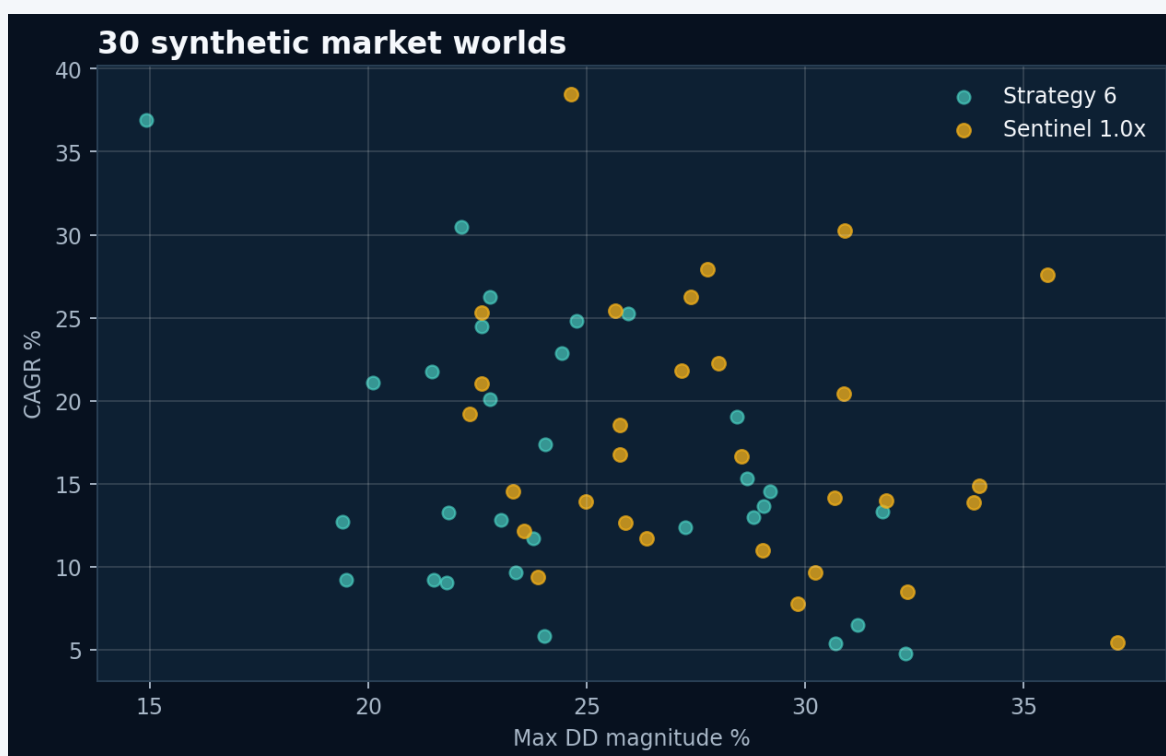
Family	CAGR win	Median CAGR Δ	Median DD Δ	Defensive
Slow Bear	83%	+1.38 pp	-6.65 pp	3.7%
False Recoveries	94%	+1.56 pp	-4.99 pp	4.3%
Breadth Rot	100%	+2.12 pp	-2.80 pp	2.7%
Dotcom Sector	94%	+1.60 pp	-5.26 pp	3.4%
Double Dip	94%	+2.00 pp	-4.88 pp	4.5%
Stagflation	89%	+1.75 pp	-7.16 pp	2.8%

Across six adversarial families and 108 worlds, Sentinel usually improved CAGR but often accepted deeper drawdown. Slow bear and stagflation are the clearest weakness. The edge is economically coherent: Sentinel pays less for ordinary-stress insurance, so it compounds more often but can remain exposed longer in grinding deterioration.

17 • OUT-OF-SAMPLE STRUCTURE



Synthetic wind-tunnel results



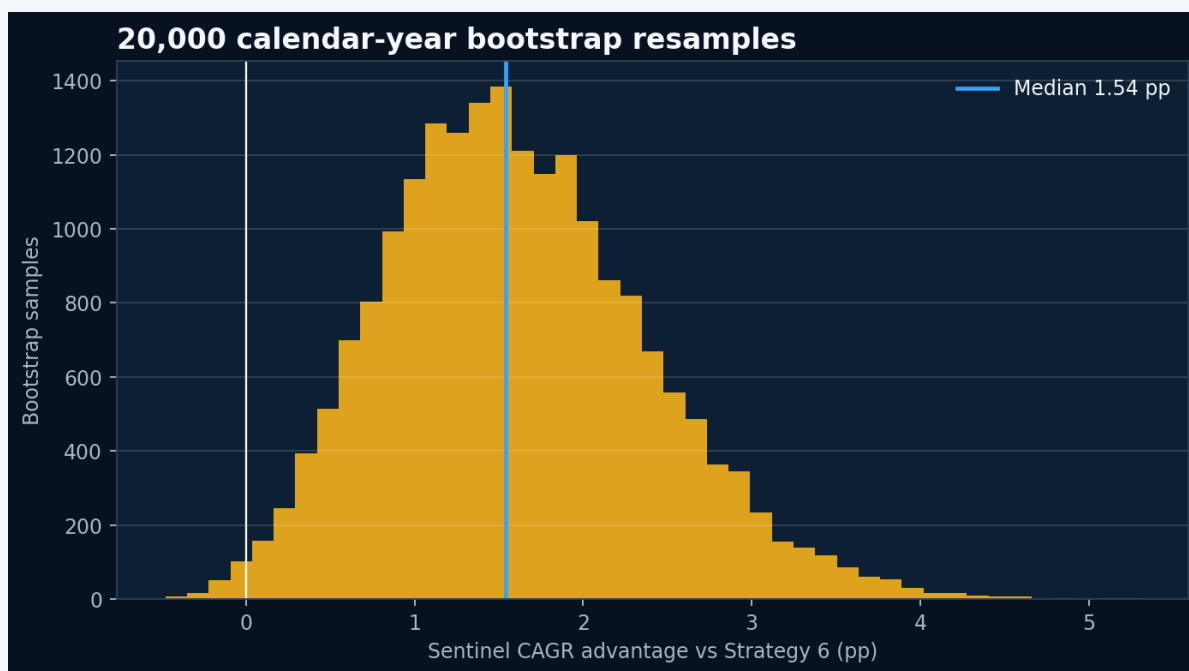
Measure	Result
Worlds	30
Higher CAGR vs Strategy 6	96.7%
Smaller DD vs Strategy 6	20.0%
Median CAGR advantage	1.64 pp
Median DD change	-3.92 pp

The asymmetry is the expected signature of removing an overactive hedge: return advantage is common, while drawdown improvement is not.

18 • 20,000 ALTERNATIVE YEAR SEQUENCES



Monte Carlo / resampling



Percentile	CAGR advantage	Terminal wealth advantage
5th	0.43 pp	7.6%
25th	1.05 pp	19.4%
50th	1.54 pp	29.4%
75th	2.10 pp	41.6%
95th	2.97 pp	62.4%

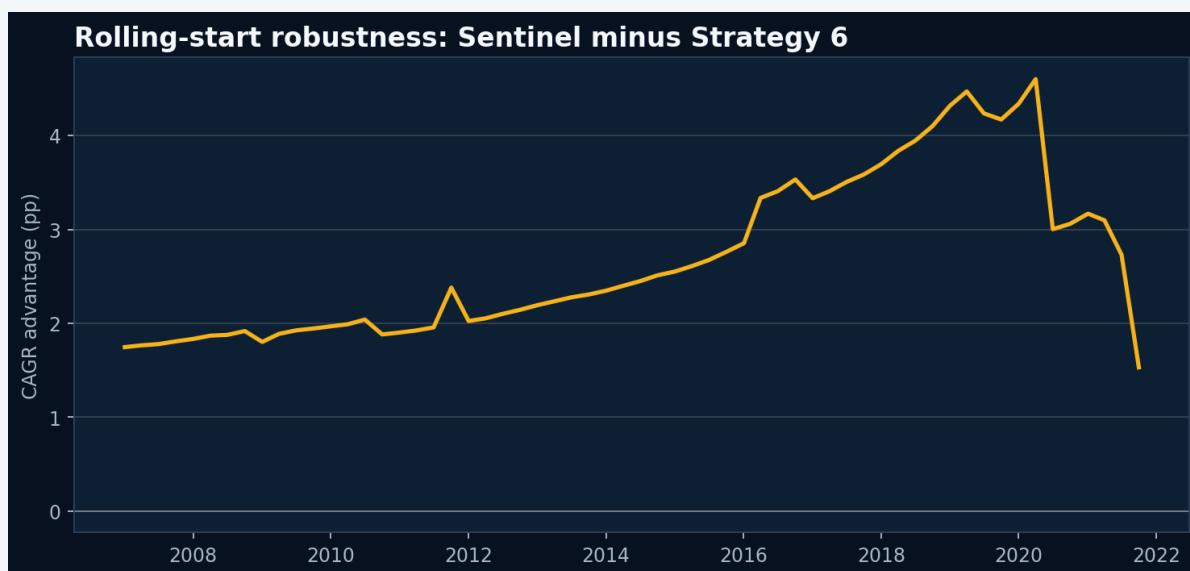
Sentinel produced greater terminal wealth than Strategy 6 in 99.28% of 20,000 calendar-year bootstrap samples. Median CAGR advantage was 1.54 percentage points.

This is a resampling robustness test, not a forecast distribution. It cannot create regimes absent from history; the adversarial and synthetic suites address that limitation.

19 • IS THE RESULT DEPENDENT ON ONE LUCKY SLICE?



Robustness suite



Rolling-start validation contains 60 alternative starts. Median Sentinel CAGR advantage is 2.39 percentage points; terminal wealth is higher in 100.0% of tested starts.

Block	S6 CAGR	Sentinel CAGR	CAGR Δ	DD Δ
2007-2010	23.4%	24.5%	+1.1 pp	+1.1 pp
2011-2014	16.8%	16.9%	+0.1 pp	-3.3 pp
2015-2018	8.6%	8.3%	-0.3 pp	-4.9 pp
2019-2022	19.5%	24.9%	+5.4 pp	+4.8 pp
2023-2026	39.4%	42.4%	+3.0 pp	+0.8 pp

20 • WHAT CAN BE REMOVED WITHOUT BREAKING THE SYSTEM?



Ablation and simplification

Crisis replaced by S6	CAGR change	DD change	Wealth change
GFC	-0.09 pp	+0.00 pp	-1.4%
2011	-0.02 pp	+0.69 pp	-0.3%
Q4_2018	+0.00 pp	+0.00 pp	+0.0%
COVID	-0.46 pp	+0.00 pp	-7.3%
2022	+0.11 pp	+0.00 pp	+1.7%

Leave-one-crisis-out testing shows no single crisis explains the result. COVID contributes materially, but replacing that episode with Strategy 6 behavior does not erase Sentinel's overall historical advantage.

Simplification proof: the one-actuator / two-detector architecture produced the same severe sessions, allocations, transition dates and values, episode boundaries and reconstructed NAV, with zero differing next-open allocation sessions.

21 • USEFUL CRITIC, WEAK ACTUATOR



LLM-in-the-loop finding

Exact next-open policy	CAGR	Max DD	Ending multiple
Strategy 6	20.42%	-28.01%	41.08x
Sentinel 1.0x	22.12%	-23.93%	54.42x
Sentinel + shadow LLM HIGH→40%	21.54%	-22.99%	49.48x

A shadow-LLM HIGH signal restoring 40% defense improved maximum drawdown by about 0.94 percentage points but reduced CAGR by about 0.58 percentage points. An aggressive HIGH-to-zero intervention sacrificed more CAGR without further maximum-drawdown benefit.

Canonical Sentinel 1.0x gives the LLM no trading authority. The LLM may act as critic, risk annotator or research hypothesis generator only.

22 • WHAT CAN STILL GO WRONG



Failure modes and safeguards

Failure mode	Required safeguard
Look-ahead	Old allocation owns overnight; new begins at actual open
Shadow contamination	Immutable full-exposure shadow
Survivorship / ID drift	Point-in-time eligibility + permanent IDs
Corporate-action divergence	Identical live/shadow semantics
Missing marks	Block admissions; never invent equity
Missing transition open	Explicit fill policy + coverage report
Threshold drift	Hash and version every rule
Restart divergence	Atomic persistence + parity tests
Overfitted machinery	No dynamic threshold tuning
Slow-bear exposure	Accepted weakness; persistent detector
Defensive instrument mismatch	Validate production mapping separately
LLM narrative overreaction	No allocation authority

23 • A BUILD IS SENTINEL ONLY IF THESE PASS



Certification acceptance tests

- 01 Golden Wealth Core parity across signals, ordering, eligibility, state, corporate actions, marks, terminal settlements and ledger hashes.
- 02 Same target portfolio/pending actions across live engine, wind tunnel and backtester for identical session state and data.
- 03 Ordinary stress at exactly -15.5% changes memory only; target remains 1.00.
- 04 Persistent boundaries: age 30 inclusive; anchor return $\leq -2\%$; $r40 \leq -3\%$; damaged $\geq 75\%$; green $\leq 25\%$.
- 05 Persistent recovery: min 20 sessions; five healthy confirmations; $r20$ strictly >0 ; damaged $\leq 60\%$; green $\geq 20\%$.
- 06 Fast systemic boundaries and recovery match frozen systemic source exactly.
- 07 Close-t decision executes open-t+1; overnight belongs to old allocation; cost only on changed notional.
- 08 Shadow remains fully active during severe stress and survives restart without state loss.
- 09 Corporate-action parity live vs shadow, including conversions, delistings, dividends and terminal settlements.
- 10 Missing-price and unresolved-equity policies are explicit, logged and deterministic.
- 11 Configuration, source, data-lineage and result hashes embedded in every certified replay package.
- 12 Reproduce the exact-next-open reference metrics within agreed numerical tolerance before promotion.

24 • SENTINEL 1.0X RESEARCH REFERENCE



Frozen configuration

```
strategy_id: sentinel_1_0x_research_v1
wealth_core:
  maximum_positions: 25
  nominal_entry_weight: 0.04
  max_new_entries_per_session: 1
  top_fraction: 0.10
  minimum_leadership_population: 25
  momentum_lookback: 126
  skip_recent_sessions: 21
  freshness_minimum: 0.00
  volatility_profile: log_returns_certified_v1
  review_age_sessions: 119
  trailing_stop_retention: 0.70
  cooldown_sessions: 21
  transaction_cost_bps: 10
sentinel:
  ordinary_stress_drawdown: -0.155
  ordinary_target_core: 1.00
  severe_target_core: 0.00
  slow_entry:
    minimum_stress_sessions: 30
    max_return_since_anchor: -0.02
    max_shadow_return_40: -0.03
    min_damaged_breadth: 0.75
    max_green_breadth: 0.25
  slow_recovery:
    minimum_state_sessions: 20
    confirmation_sessions: 5
    min_shadow_return_20: 0.00 # strict >
    max_damaged_breadth: 0.60
    min_green_breadth: 0.20
execution:
  decision_time: official_close
  effective_time: next_session_open
  overlay_transition_cost_bps: 10
  leverage: 1.00
defensive_asset:
  research_proxy: BIL_total_return
shadow:
  mode: immutable_full_exposure
  persistence: atomic
llm:
  trading_authority: none
```

Versioning rule: any change to a threshold, inequality, feature definition, price domain, ordering profile, defensive asset, execution timestamp, cost model, recovery rule, leverage or LLM authority creates a new immutable strategy version.

25 • WHAT THIS DOCUMENT IS - AND IS NOT



Research record and disclosures

Implementation source: Wealth Core modules from the repository archive used in this research, including signals, engine, state, eligibility, marks, terminal, adapter, execution model and risk profile.

Result record: exact-next-open paths, crisis tables, rolling starts, calendar blocks, leave-one-crisis-out, 30 synthetic worlds, 108 adversarial worlds, 20,000 year-bootstrap resamples, drawdown-tail analysis, simplification-equivalence proof and LLM shadow experiment.

Status: Sentinel 1.0x is the canonical research champion architecture. Repository implementation and promotion testing are still required before the overlay should be called a certified deployed strategy.

Historical CAGR is not a forecast. Adversarial testing explicitly shows regimes where Sentinel accepts deeper drawdown than Strategy 6. Changing point-in-time data, corporate-action treatment, price domains or execution semantics can change results.

Design conclusion: simplicity is intentional. Additional controllers, fuses and narrative overrides generally bought small amounts of drawdown protection at the cost of compounding. Sentinel retains only the complexity that demonstrated structural value.

SENTINEL 1.0x

Own durable winners. Let ordinary corrections happen.

Act only when severe stress is earned by evidence.